

Cybersecurity Engineering Approach

Revision

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SME

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Abstract

This document describes the general approach used to undertake the various activities specified by the AVCDL.

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Overview

Product-level cybersecurity activities are typically considered in isolation. This is true of both cybersecurity practitioners and non-practitioners. It is easy to do for two reasons. First product-level cybersecurity evolved on an as-needed basis. Second, it is not widely appreciated that cybersecurity is an emergent property of the system.

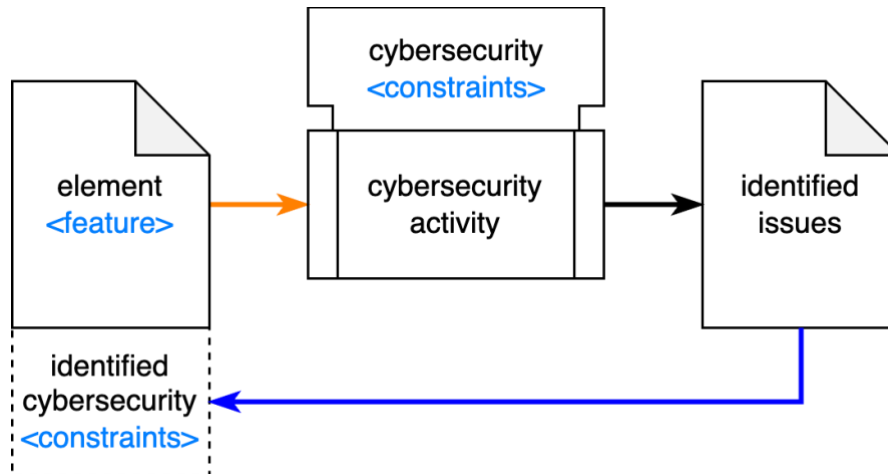
Generally speaking, product-level cybersecurity currently uses a set of individual activities to address specific concerns within a system without regard to either overall workflow or process reuse.

From the standpoint of impact, arguably the most important group of cybersecurity activities are those which directly focus on the soundness of the software within the product. These activities lend themselves to a uniform engineering approach and that's what this document will address.

Workflow

AVCDL processes apply the workflow shown below.

Note: Unlike typical AVCDL process workflow diagrams, the actor, activity feedback, and report elements are excluded in order to focus on the topic at hand.



The cybersecurity activity takes as its input a **feature** of the **element**. This feature represents some aspect of the element which may be reasoned on for the activity. It may be a design, source code, functional requirements, or any other quantifiable aspect of the **element** under consideration.

The **feature** may be constrained by a set of **identified cybersecurity constraints**. These may be design principles, cybersecurity requirements, or best practices.

The **cybersecurity activity** could be requirements analysis, attack surface analysis, threat modeling, static analysis, dynamic analysis, secure code review, or any other appropriate activity.

The **cybersecurity activity** is governed, generally speaking, by a particular set of **cybersecurity constraints**.

These **cybersecurity constraints** are a superset of the **identified cybersecurity constraints** attached to the particular feature of the **element** under consideration.

The **cybersecurity activity** will result in a set of **identified issues** being created.

These **identified issues** need to point to specific **cybersecurity constraints** that either need to be attached to the **element** or be verified as actually having been applied to the **element**.

Note: It is not uncommon to have a constraint you know needs to be there and it doesn't happen to be in use.

Element Features / Cybersecurity Constraints

The following table details the element features and cybersecurity constraints applicable to various cybersecurity activities.

Note: Only those AVCDL processes most directly focused on the soundness of the product's software are included.

cybersecurity activity	element feature	cybersecurity constraint
requirements analysis ^[2]	functional requirements	requirements
design analysis ^[3]	design	design principles
attack surface analysis ^[4]	ports / protocols / processes	excess exposure
threat model analysis ^[5]	data flow diagrams	requirements
secure code review ^[6]	source code	development practices
static analysis ^[7]	source code	sound code practices
dynamic analysis ^[8]	compiled code	sound code practices
fuzz testing ^[9]	instrumented code	sound code practices
penetration testing ^[10]	operational system	see note below

Note: Penetration testing is an *ad hoc* activity where the testers utilize available information to attempt to induce failure states within an operational system. There is no established set of cybersecurity constraints by which this activity is governed. It has neither an industry (or regulatory) agreed upon set of defined outcomes nor metrics. It is included here for completeness.

Additional information regarding the specific element features and cybersecurity constraints for a given cybersecurity activity can be found within the applicable AVCDL secondary and tertiary documents.

References

1. **AVCDL** (AVCDL primary document)
2. **Element-level Security Requirements** (AVCDL secondary document)
3. **Security Design Review Report** (AVCDL secondary document)
4. **Attack Surface Analysis Report** (AVCDL template)
5. **Threat Modeling Report** (AVCDL elaboration document)
6. **Secure Code Review Summary** (AVCDL secondary document)
7. **Static Analysis Report** (AVCDL tertiary document)
8. **Dynamic Analysis Report** (AVCDL video)
9. **Fuzz Testing Report** (AVCDL secondary document)
10. **Penetration Testing Report** (AVCDL working material)